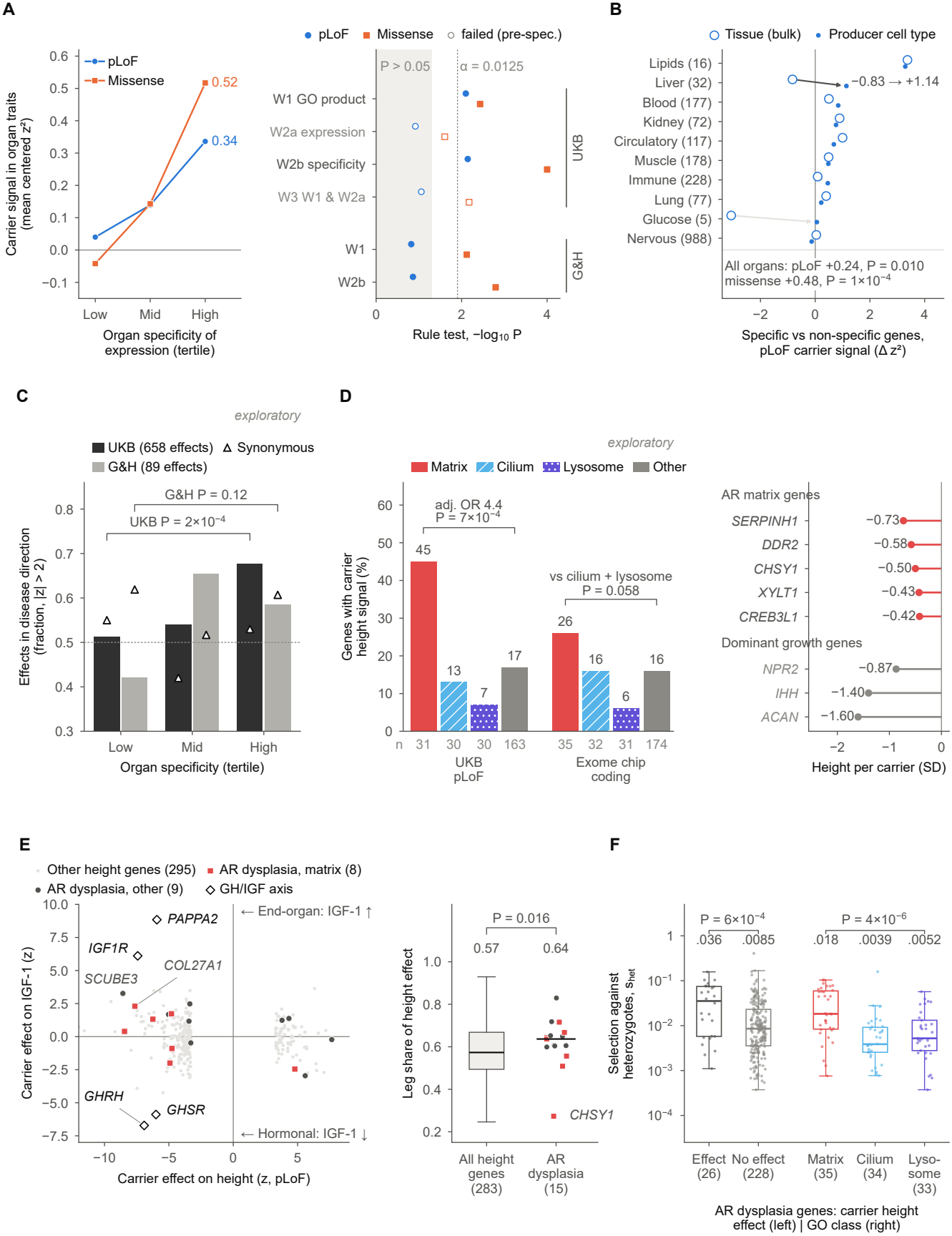

A
pLoF
Missense
0.52
0.34
Carrier signal in organ traits (mean centered z²)
Low
Mid
High
Organ specificity of expression (tertile)
pLoF
Missense
failed (pre-spec.)
P > 0.05
α = 0.0125
W1 GO product
W2a expression
W2b specificity
W3 W1 & W2a
W1
W2b
UKB
G&H
Rule test, −log10 P
B
Tissue (bulk)
Producer cell type
Lipids (16)
Liver (32)
Blood (177)
Kidney (72)
Circulatory (117)
Muscle (178)
Immune (228)
Lung (77)
Glucose (5)
Nervous (988)
−0.83 → +1.14
All organs: pLoF +0.24, P = 0.010
missense +0.48, P = 1×10⁻⁴
Specific vs non-specific genes, pLoF carrier signal (Δ z²)
C
exploratory
UKB (658 effects)
Synonymous
G&H (89 effects)
G&H P = 0.12
UKB P = 2×10⁻⁴
Effects in disease direction (fraction, |z| > 2)
Low
Mid
High
Organ specificity (tertile)
D
exploratory
Matrix
Cilium
Lysosome
Other
adj. OR 4.4
P = 7×10⁻⁴
vs cilium + lysosome
P = 0.058
Genes with carrier height signal (%)
45
13
7
17
26
16
6
16
n 31 30 30 163
35 32 31 174
UKB pLoF
Exome chip coding
AR matrix genes
SERPINH1 −0.73
DDR2 −0.58
CHSY1 −0.50
XYLT1 −0.43
CREB3L1 −0.42
Dominant growth genes
NPR2 −0.87
IHH −1.40
ACAN −1.60
Height per carrier (SD)
E
Other height genes (295)
AR dysplasia, matrix (8)
AR dysplasia, other (9)
GH/IGF axis
PAPPA2
IGF1R
COL27A1
SCUBE3
GHRH
GHSR
← End-organ: IGF-1 ↑
← Hormonal: IGF-1 ↓
Carrier effect on IGF-1 (z)
Carrier effect on height (z, pLoF)
P = 0.016
0.57
0.64
CHSY1
Leg share of height effect
All height genes (283)
AR dysplasia (15)
F
P = 6×10⁻⁴
.036
.0085
P = 4×10⁻⁶
.018
.0039
.0052
Selection against heterozygotes, s_het
Effect (26)
No effect (228)
Matrix (35)
Cilium (34)
Lyso-some (33)
AR dysplasia genes: carrier height effect (left) | GO class (right)